

ZHAW – Zurich University of Applied Sciences

School of Engineering

Drone Control Tuning

Development of an Offline Betaflight Drone Control Tuner

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1 Abstract

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2 Foreword

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3 Introduction

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3.1 Initial situation

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3.2 Task and Objective

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4 Preparation

4.1 Chirp

4.1.1 Chirp Concept

A chirp is an input signal that sweeps continuously over a defined range (e.g. from a low frequency to a high frequency). This signal excites the system on multiple frequencies in one run and thus lets you observe the frequency response and step response, the foundation of controller tuning. Mathematically, a chirp signal can be represented as a sinusoidal wave with instantaneous frequency that varies linearly or exponential with time. For more information about the mathematical background, please have a look in the document [Chirp Signal](#).

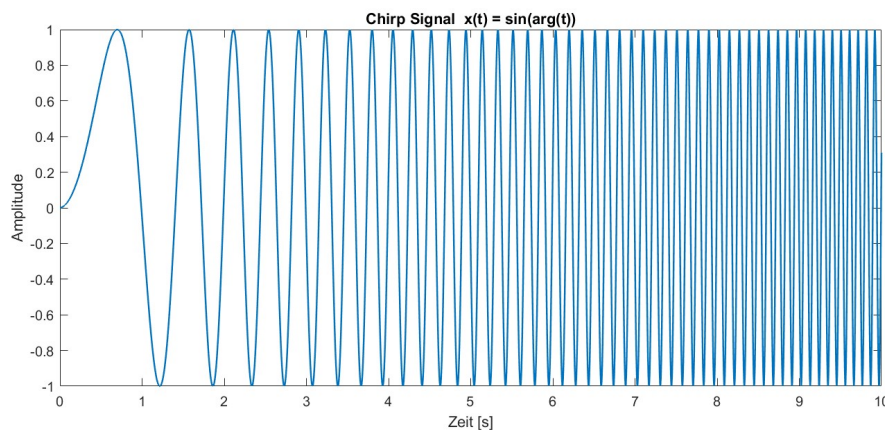


Abbildung 1: Chirp Signal example

4.1.2 Chirp on a drone

In the context of tuning a drone's flight controller, an exponential sweep is used. This is due to the dominant low-frequency dynamics, where low frequencies exhibit more predictable responses, allowing the sweep to allocate more time to higher frequencies for better resolution of system behavior. Additionally, multirotor vibration resonances are typically spaced logarithmically, making exponential sweeps more suited for efficient excitation across the frequency range.

The process involves these steps:

1. **Signal Injection:** The chirp signal is added to the pilot's stick input. This composite signal becomes the setpoint for the PID controller.
2. **System Response:** The drone attempts to follow this rapidly changing setpoint. The motors react, and the drone will oscillate at varying frequencies corresponding to the chirp signal.
3. **Data Logging:** While the chirp is active, high-resolution data from the gyroscope and the motor outputs are recorded using Betaflight's Blackbox feature. This data captures how the drone responds to the control inputs at different frequencies.

The recorded log file, containing the input chirp and the corresponding gyroscope output, is then used for offline analysis to determine the frequency response of the system.

4.1.3 Chirp programming

The implementation of the chirp signal in Betaflight is straightforward. You call the function `chirpInit()` and pass:

- a struct of type `chirp_t` defined in `chirp.h`
- starting frequency
- end frequency
- signal length
- loop-time in microseconds

This initialises the values in the data struct and:

- converts loop period from microseconds to seconds
- calculates the number of samples in the signal duration
- calculates the exponential growth factor
- precomputes the constants used to get the phase from the frequency evolution
- calls `chirpReset()` function

The `chirpReset()` function sets:

- internal counter to 0
- bool `isFinished` to false

then calls the function `chirpResetSignals()` to set signal-related fields to 0

Once the chirp is initialised use the `chirpUpdate()` function once per loop iteration. This function checks:

- if bool `isFinished` is true -> return false
- else if internal counter is equal to the number of samples inside the chirps period
-> set bool `isFinished` true, call `chirpResetSignals()`, return false

if none of the above is true the ...